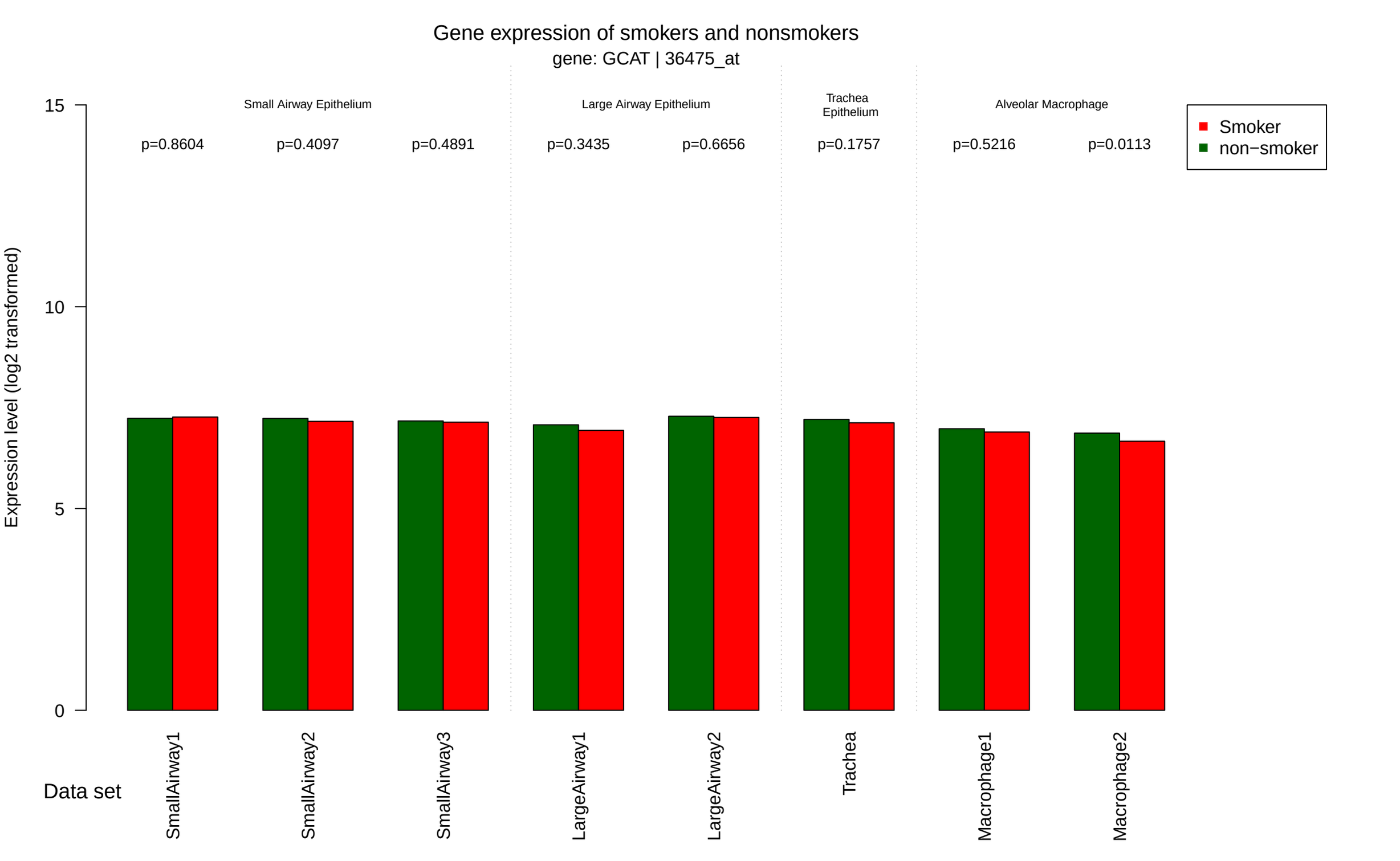
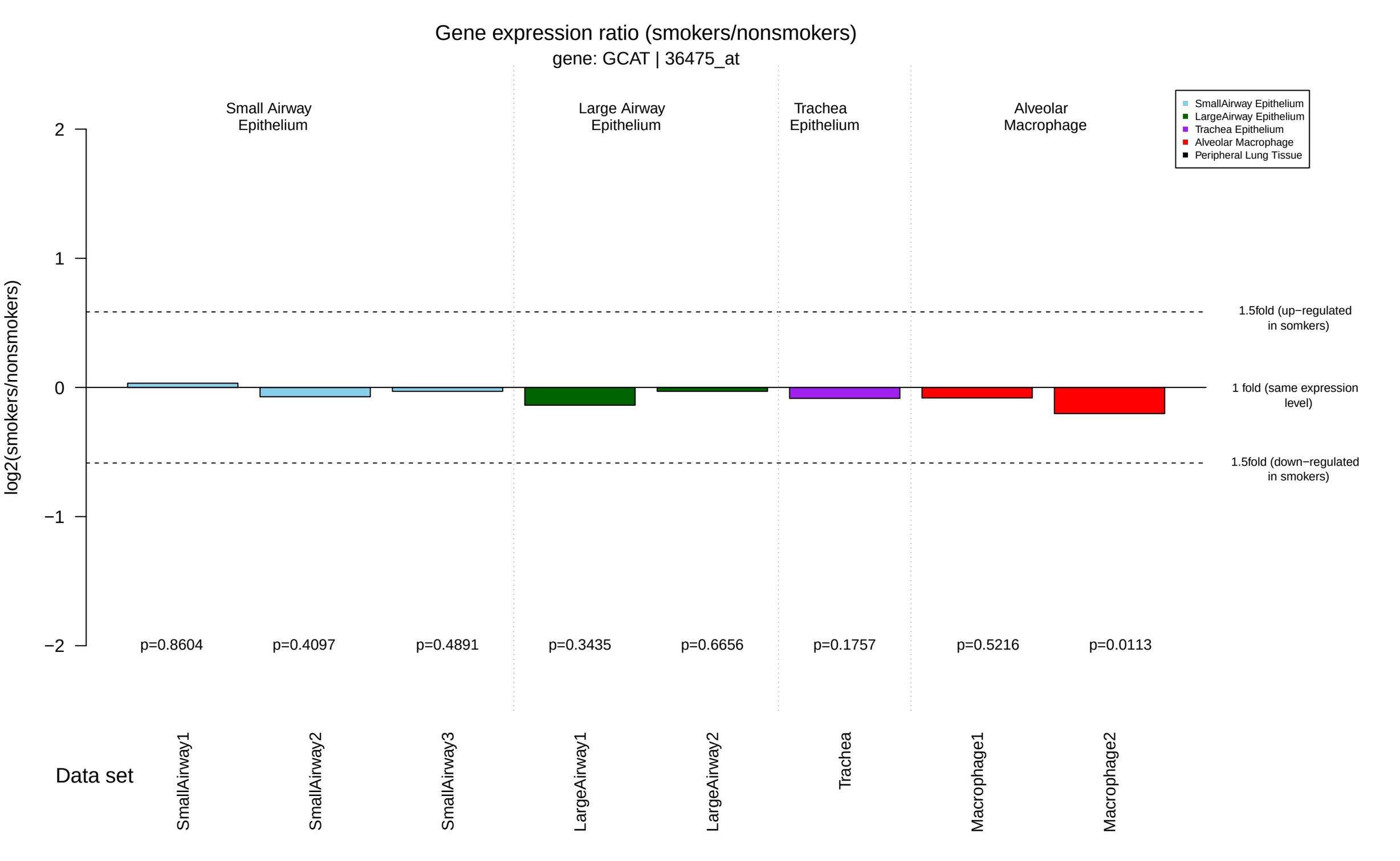
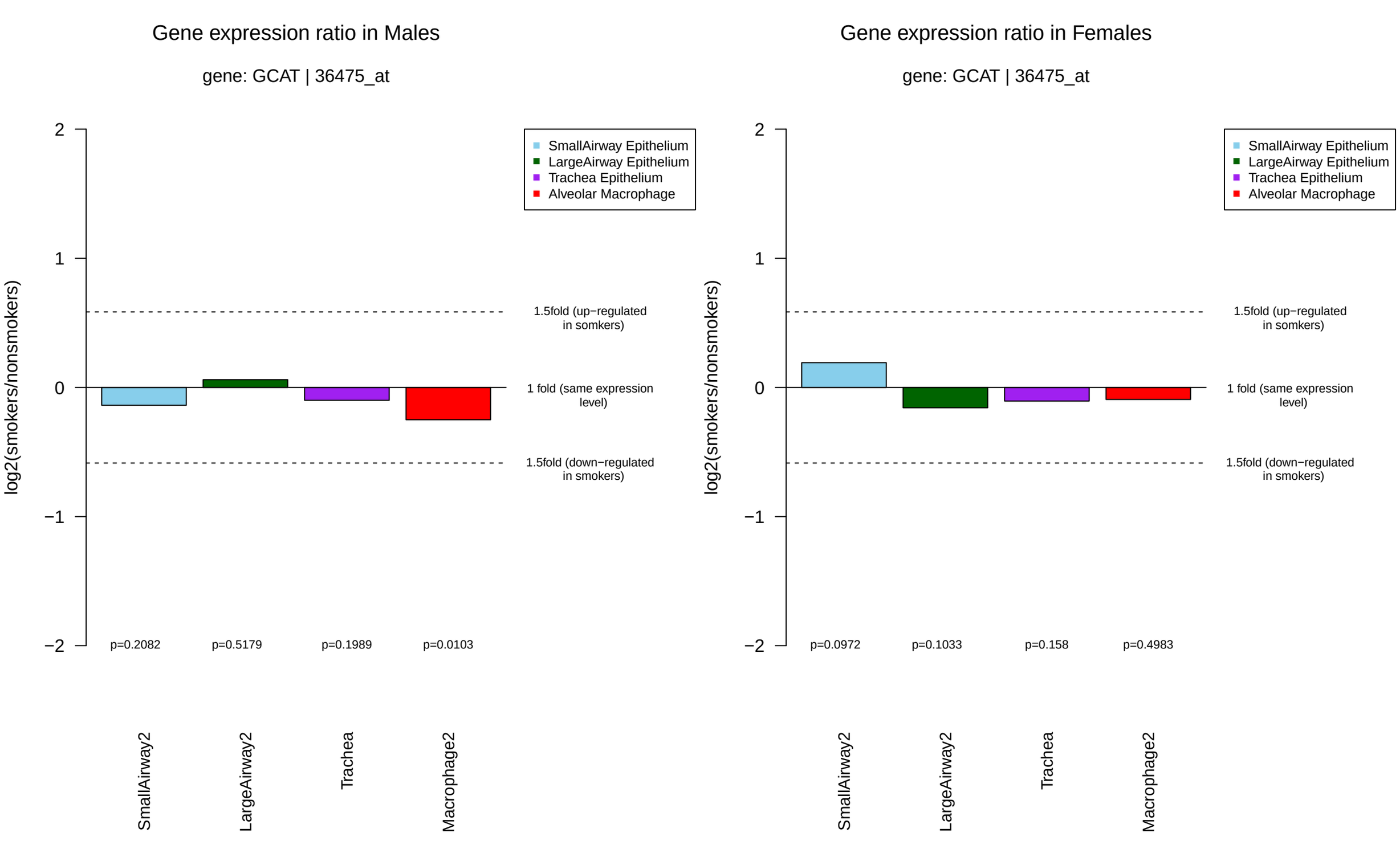
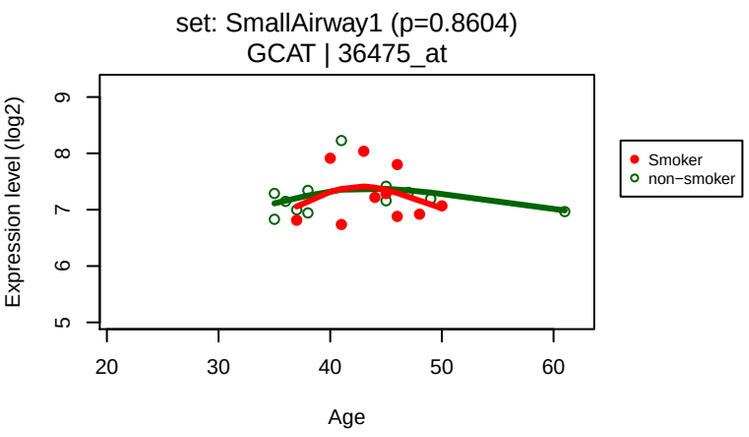
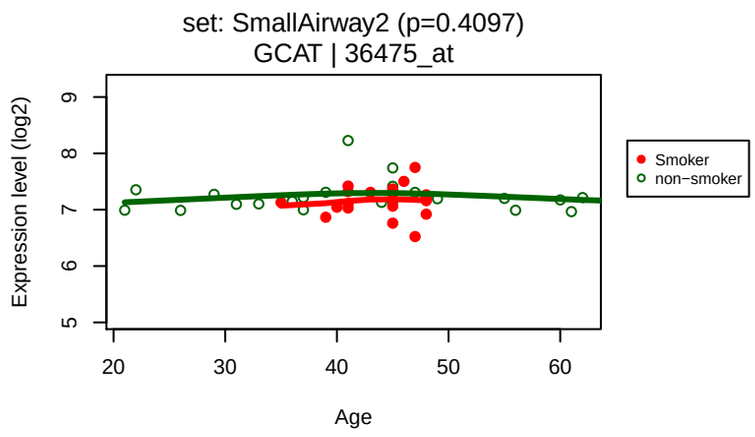
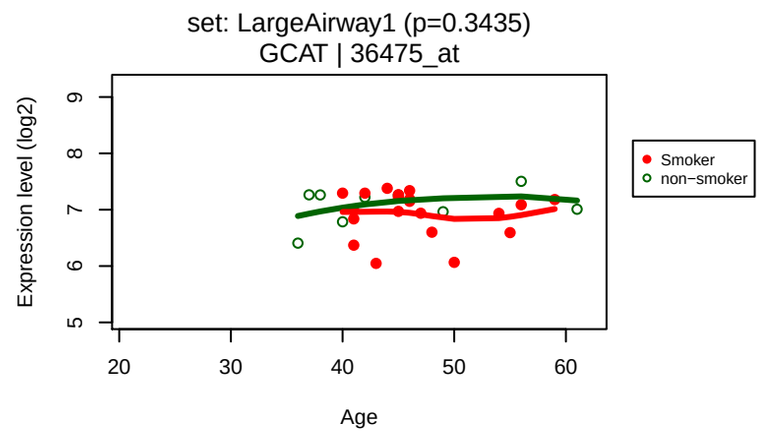

Gene expression of smokers and nonsmokers
gene: GCAT | 36475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8604
p=0.4097
p=0.4891
p=0.3435
p=0.6656
p=0.1757
p=0.5216
p=0.0113
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

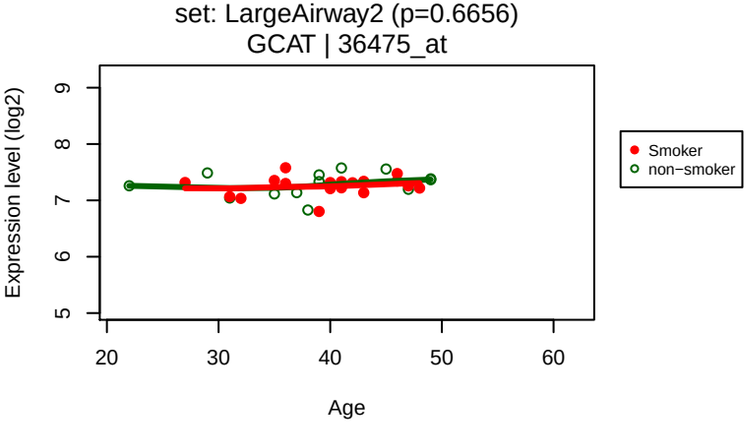

Gene expression ratio (smokers/nonsmokers)
gene: GCAT | 36475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8604
p=0.4097
p=0.4891
p=0.3435
p=0.6656
p=0.1757
p=0.5216
p=0.0113
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

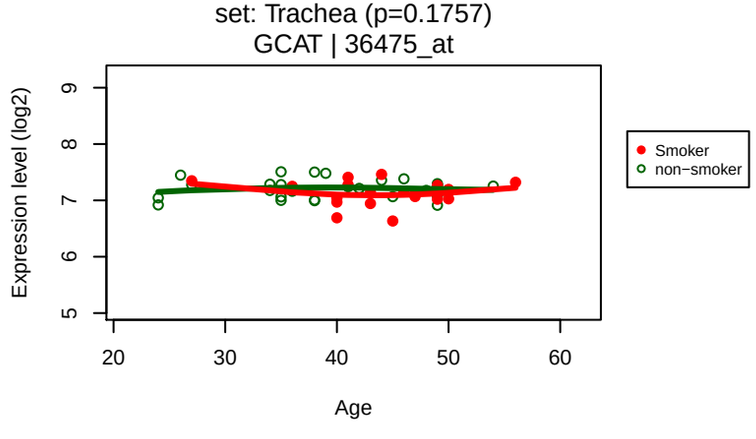

Gene expression ratio in Males
gene: GCAT | 36475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2082
p=0.5179
p=0.1989
p=0.0103
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCAT | 36475_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0972
p=0.1033
p=0.158
p=0.4983
SmallAirway2
LargeAirway2
Trachea
Macrophage2

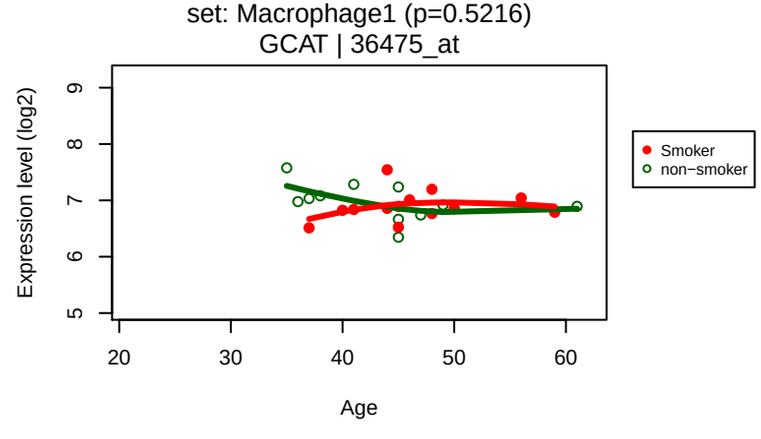

set: SmallAirway1 (p=0.8604)
GCAT | 36475_at
Expression level (log2)
Age
Smoker
non−smoker

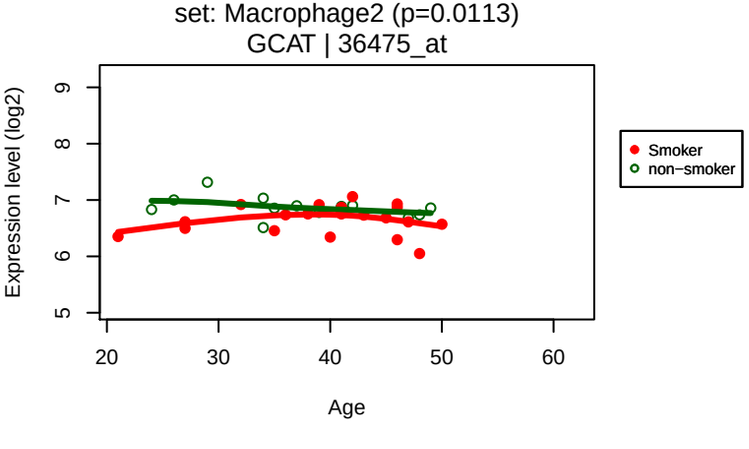

set: SmallAirway2 (p=0.4097)
GCAT | 36475_at
Expression level (log2)
Age
Smoker
non−smoker

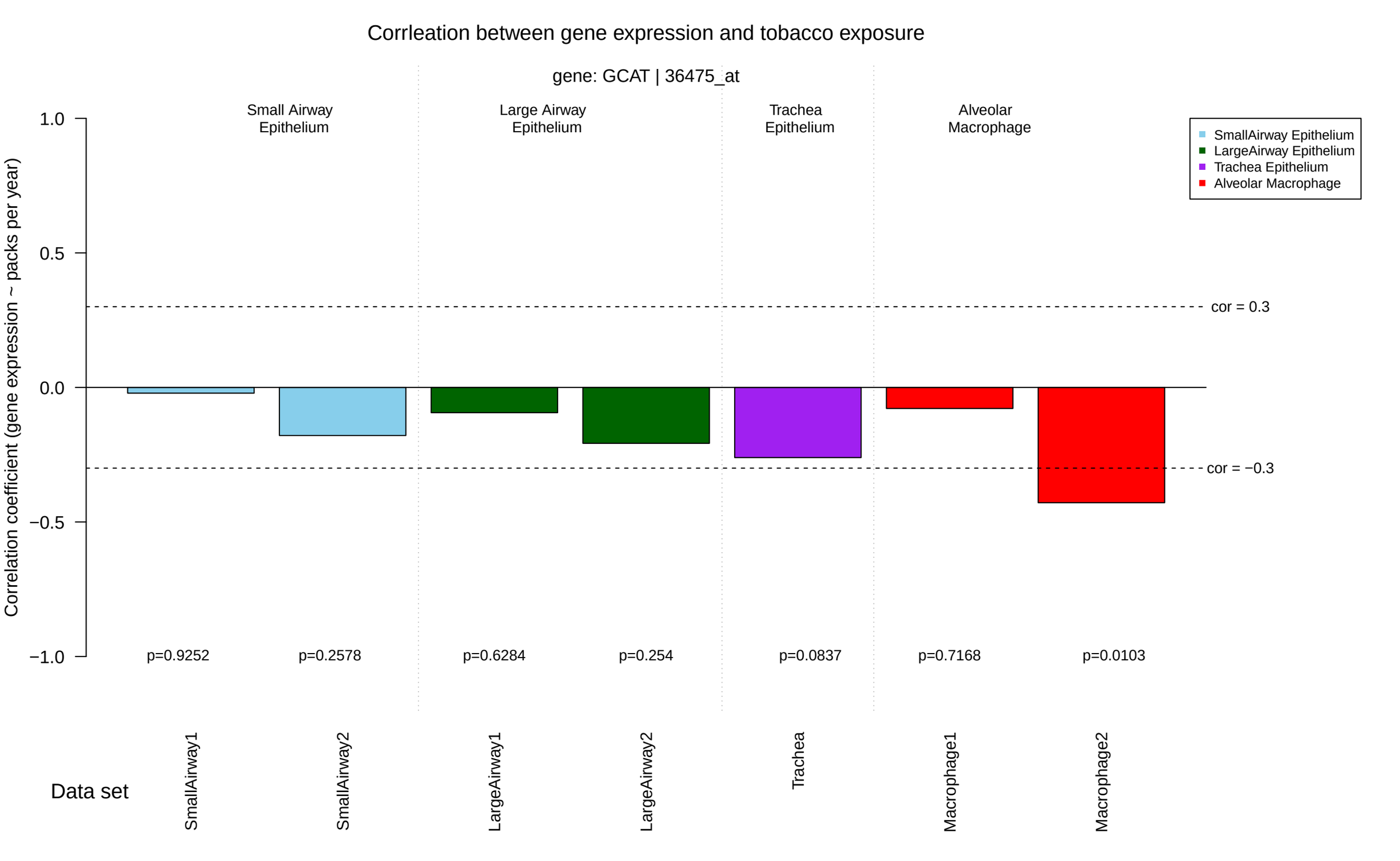

set: LargeAirway1 (p=0.3435)
GCAT | 36475_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6656)
GCAT | 36475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1757)
GCAT | 36475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5216)
GCAT | 36475_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0113)
GCAT | 36475_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCAT | 36475_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9252
p=0.2578
p=0.6284
p=0.254
p=0.0837
p=0.7168
p=0.0103
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2